EB-JEPA for EEG Time Series

Self-Supervised Abnormality Detection on TUAB

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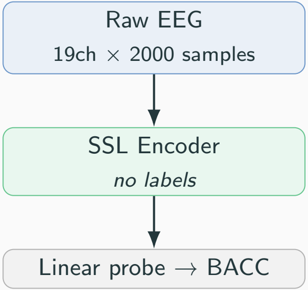
Task & Motivation

TUAB — TUH Abnormal EEG Corpus

- Binary: normal / abnormal clinical EEG
- 19 channels, 200 Hz, 10 s windows
- 2717 train / 276 eval recordings
- Patient-disjoint split — no leakage

Challenge: learn a transferable EEG representation *without* labels

- Labelling EEG requires licensed neurologists
- Unlabelled data is abundant
- Can self-supervised learning close the gap?



Our Approach — EB-JEPA Two-View SSL

Energy-Based JEPA (two-view invariance):

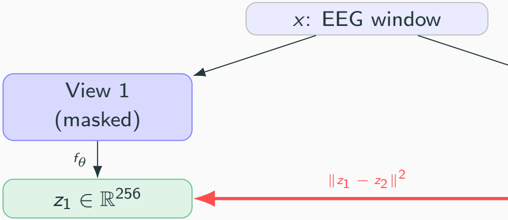
- Take one EEG window; create two *augmented views*
- Push both through the same encoder f_θ
- Force $z_1 \approx z_2$ while preventing collapse

Augmentations:

- Gaussian noise, amplitude jitter, channel drop
- **+ Corruption** (our design): 20% time masking, $\pm 6\sigma$ spike injection

What we contributed:

- 4 encoder architectures
- 2 SSL objectives (VICReg / SIGReg)
- Full ablation + bug diagnosis



Four Encoder Architectures

Encoder	Tokenisation	Architecture	Params
Conv	Strided Conv1D (k=7)	4 BN+GELU blocks	0.4M
LaBraM	1s patches per channel	ViT (190 tokens)	1.2M
BIOT	FFT magnitude patches	ViT (190 tokens)	1.2M
EEGPT	Cross-ch pool → 10 tokens	Temporal transformer	0.8M

VICReg loss (Bardes et al. 2022):

$$\mathcal{L} = 25 \underbrace{\|z_1 - z_2\|^2}_{\text{invariance}} + 25\,V + 1\,C$$

Hinge variance + off-diagonal covariance penalty.

SIGReg / BCS: Replaces $V+C$ with Epps–Pulley Gaussianity test on random projections. Stronger anti-collapse, less hyperparameter-sensitive.

The Evaluation Trap

Per-window vs Per-recording — 5 pp difference, same encoder

- **Literature** (BIOT, LaBraM, FEMBA, EEGPT): reports **per-window** — each 10 s segment is a separate test point. Up to 409 455 test samples from 2339 recordings.
- **Our evaluation**: mean-pools 16 windows per patient into one embedding → **per-recording** score.
- Per-recording is ≈5 pp *higher* than per-window for the same encoder (noise averaging).

Our initial mistake
We reported per-recording BACC **0.836** against LaBraM's per-window BACC **0.814** and concluded "we match LaBraM." That comparison is invalid.

Correction: our best *per-window* score is **0.775** — below BIOT (0.802) and LaBraM (0.814).

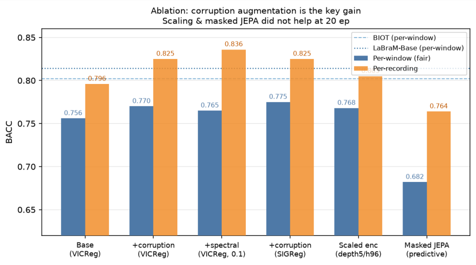
Note: per-recording is arguably the right *clinical* metric (you classify patients, not 10s snippets), but it is not what the benchmark measures.

Results — Fair Per-Window Comparison

Method	BACC	AUROC
ContraWR	0.775	0.846
BIOT pretrained	0.802	0.874
LaBraM-Base	0.814	0.902
EB-JEPA base	0.756	0.845
+ **corruption**	**0.770**	0.848
+ **SIGReg** + **corrupt**	**0.775**	**0.856**
EEGNet (supervised)	0.796	0.882

✓ Matches ContraWR — competitive SSL baseline

× Still 4 pp below LaBraM



Ablation — What Actually Moved the Needle

Variant	ΔBACC	ΔAUROC
Base VICReg	—	—
+corruption	+0.014	+0.003
+spectral (0.1)	+0.009	+0.004
SIGReg +corrupt	+0.019	+0.011
Scaled encoder (depth 5)	≈ 0	—
150-epoch pretraining	≈ 0	—
Masked-prediction JEPA	−0.074	—
Multi-corpus 4× data	+0.012	—

Key lesson:

- Corruption augmentation is the dominant win
- SIGReg $\geq$ VICReg (consistent with paper)
- Scaling & more data did *not* help — binding constraint is **encoder capacity**
- Frame-prediction JEPA underperformed with global-pool probe

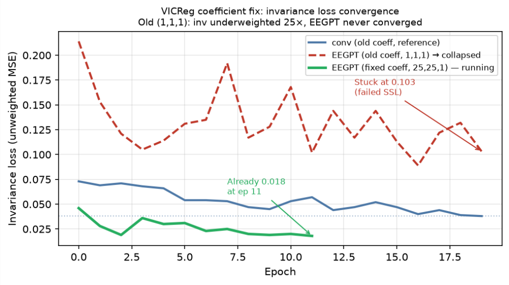
Failure Analysis — VICReg Bug + EEGPT Collapse

Bug: wrong VICReg coefficients

- Implementation used $(1, 1, 1)$ instead of $(25, 25, 1)$
- Invariance underweighted **25**$\times$ — encoder barely penalised for mismatching views
- Fix: added `inv_coeff=25` to `VICRegLoss`

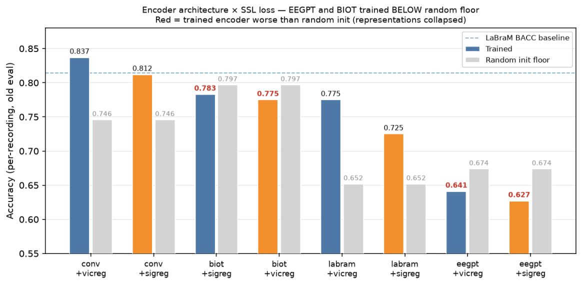
EEGPT collapse (two causes)

1. **Architectural**: 19 channels $\rightarrow$ 1 pooled token per patch. Only 10 temporal tokens total. Channel-drop augmentation makes the two views structurally incompatible *before* the transformer sees them.
2. **Underweighted invariance** (the bug above): optimizer never forced the encoder to fix this.



Fixed run (green): inv_loss 0.018 at epoch 11
Old run (red dashed): stuck at 0.103 at epoch 19

Encoder Ranking & Collapse Diagnosis



Observations:

- **Conv** (simple, 0.4M) beats all transformers
- LaBraM-style (raw patches) is the best transformer architecture for VICReg
- BIOT: FFT magnitude discards phase $\rightarrow$ representation captures frequency statistics, not temporal pathology patterns. Slightly below random with SIGReg.
- EEGPT: systematic collapse with old VICReg coefficients. Fix running — inv_loss already 5$\times$ lower.

Transfer to TUEV (6-class Event Classification)

Task: TUEV — 6 EEG event types (spsw, gped, pled, eyem, artf, bckg)
Protocol: freeze TUAB-pretrained encoder; train linear probe on TUEV

Encoder	BACC	Kappa
Random init	0.337	0.110
EB-JEPA (TUAB frozen)	**0.364**	**0.141**
BIOT (paper)	0.528	0.435

Takeaways:

- SSL representation *does* transfer — +0.027 BACC over random
- Still far below BIOT (0.528)
- BIOT's FFT tokenisation may be better for TUEV's frequency-based event types
- Pretraining task (binary pathology) $\neq$ transfer task (6 event types)

Conclusion

What works

- Corruption augmentation: $+0.014$ BACC (dominant gain)
- SIGReg $\geq$ VICReg with correct $(25, 25, 1)$ coefficients
- Simple Conv encoder beats all transformer variants ($3\times$ smaller)
- Frozen probe BACC 0.775 matches ContraWR
- Per-recording BACC 0.836 — competitive clinical score

What doesn't

- Scaling encoder: bottleneck is architecture, not width/depth
- EEGPT: channel-pool + channel-drop = incompatible
- Masked-prediction JEPA (20 ep, global-pool probe)
- FFT tokenisation (BIOT): discards phase/temporal dynamics

Honest position (per-window, fair to literature)
0.775 — below BIOT (0.802) and LaBraM (0.814).

Competitive with supervised EEGNet (0.796) as a *frozen* representation.

The binding constraint is encoder capacity and tokenisation quality.